

Toward Near-Foldable Surface Light Sources with Ultimate Efficiency: Ultrathin Substrates Embedded with Micron-Scale Inverted Lens Arrays

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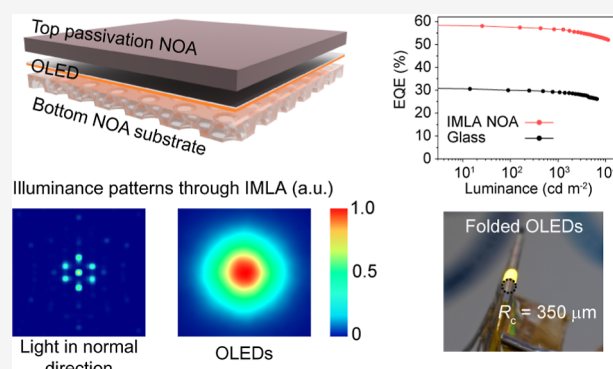
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ABSTRACT: Foldable organic light-emitting diodes (OLEDs) are essential building blocks for portable devices with expandable screens as well as more futuristic systems such as wearable or body-attachable electronic devices. Although various approaches have been proposed to realize foldable OLEDs, the efficiency enhancement techniques developed for their rigid counterparts are not always applicable due to the strict thickness limitations, making it challenging to achieve very high efficiency in a foldable OLED. Here, we propose ultrathin substrates embedded with an inverted microlens array (IMLA) as a platform on which to design and realize OLEDs that can be bent at sub-100 μm bending radius while also exhibiting very high light outcoupling efficiency. By noting the periodic arrangement of the patterns in the IMLA, the potential effects of optical diffraction on the overall emission pattern and efficiency enhancement are carefully analyzed by incorporating a bidirectional scattering distribution function via a trans-scale approach. Neutral-plane engineering is also done with finite element method simulations that examine the effect of the IMLA structures on the local modulation of the strain and stress in ultrathin devices, where the feature size of the IMLA is comparable to the overall thickness of the whole device. With the proposed method, highly efficient foldable OLEDs are demonstrated that show the maximum external quantum efficiency to be as high as 58% without optical side effects and that can withstand 10,000 trials of repeated folding cyclic tests at a bending radius of 50 μm .

KEYWORDS: foldable displays, organic light-emitting diodes, ultrathin devices, microlens arrays



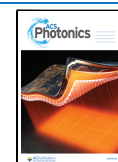
INTRODUCTION

Developing new form factors is becoming increasingly important to meet the increasing demand for both portability and a large screen in modern consumer electronics. In this regard, foldable displays have recently been developed and adopted in mobile phones, and smart personal devices with rollable screens are also on their way to commercialization.¹ The light sources in these devices should be able to deform at a bending radius (R_c) that is as small as a few millimeters or less depending on the target application.² Reducing the minimum sustainable R_c further to the sub-millimeter level is expected to extend the use of displays or light sources to more futuristic applications. For example, body-attachable devices with a good conformal coverage, as well as skin-like displays that can be integrated onto 3D objects, may be realized with an R_c value of several tens to hundreds of millimeters or less for coverage;^{3–6} light sources with the minimum R_c value as low as 5 mm or less could be covered onto pre-stretched elastomers to build stretchable displays, wherein the light sources make conformal

coverage over the wrinkles that are formed in the elastomer when it returns to the unstretched state after pre-stretching.⁷ Among several light sources, organic light-emitting diodes (OLEDs) are considered to be the ideal candidate for realizing deformable light sources that can sustain bending at such small R_c levels due to their inherently good mechanical flexibility and process compatibility with various flexible substrates. In this regard, many studies have been dedicated to realizing highly flexible OLEDs.^{8–14} In particular, those based on ultrathin substrates were found to be particularly important,^{4,12–14} as the flexural strain applied to the top of a layer deposited on top of a substrate is proportional to the ratio of the substrate

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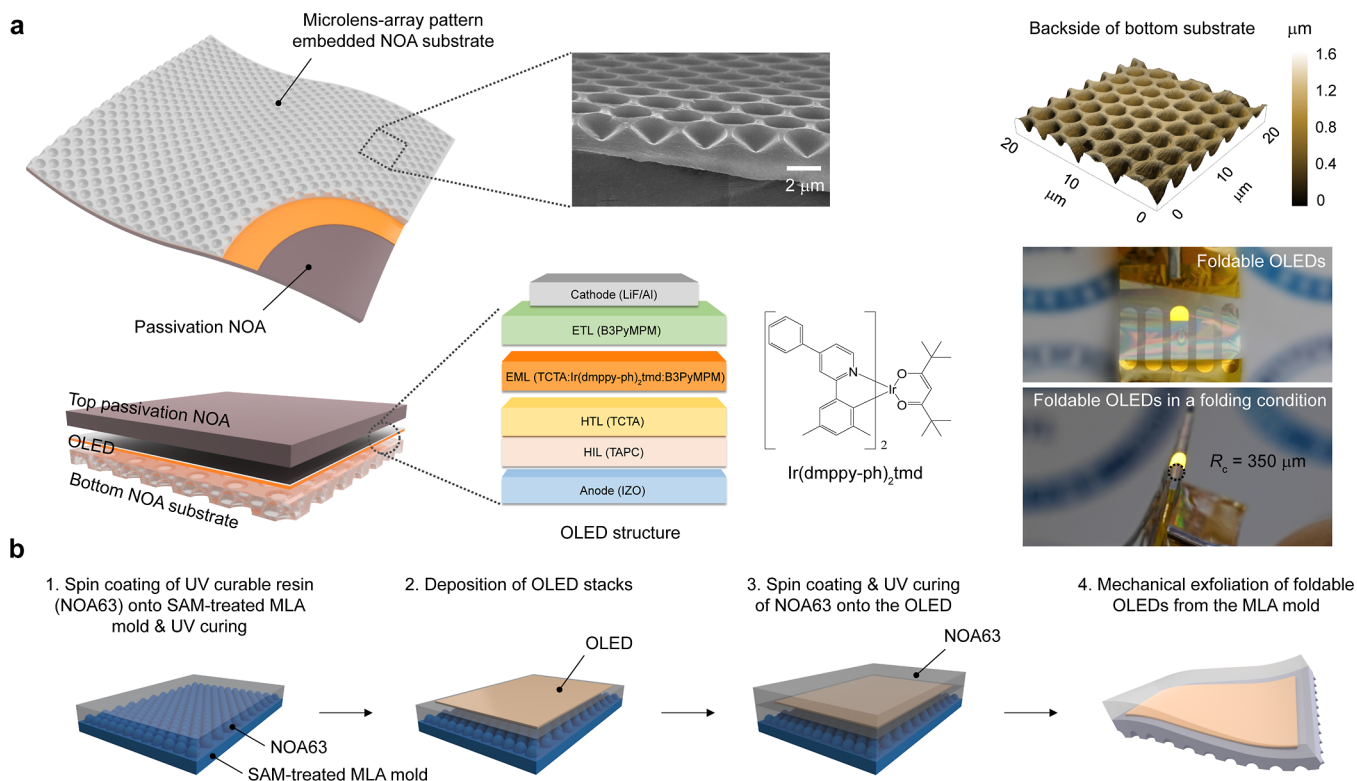


Figure 1. Schematic diagram of NOA substrates with embedded inverted microlens arrays (IMLAs) along with microscopic images and surface profile data: (a) (left) schematic diagram of the proposed foldable OLEDs based on IMLAs embedded into a NOA substrate. The inset shows a cross-sectional scanning electron microscopy (SEM) image of the IMLA-embedded NOA substrate. (Right, top) AFM topography images of the IMLA-embedded NOA substrate. (Right, bottom) Photograph of the foldable OLED in a folding condition. (b) Schematic diagram of the fabrication process used to fabricate the foldable OLED with the NOA substrates with the embedded IMLAs.

thickness (d_{sub}) to R_c and because oxides, used as transparent conductors and encapsulation materials in OLEDs, have a rather small crack-onset strain (COS) of less than ca. 1%,¹² OLEDs containing those layers are prone to bending-induced crack formation when the d_{sub} value is relatively large. Built on 6 μm thick Mylar substrates, OLEDs with oxide-based multilayer thin-film encapsulation have recently been demonstrated to show performance comparable to that of rigid control devices and operability under repeated bending cycles at an R_c value of 0.3 mm.¹²

Given such success, however, the efficiency of deformable OLEDs on ultrathin substrates has been rather limited, partly due to the fact that the light extraction techniques^{15–19} developed for rigid counterparts such as microlens arrays (MLAs) and optical scattering often involve layers that are too thick to be adopted for ultrathin OLEDs.^{13,20–23} To provide a remedy for this dilemma, we present a simple yet highly effective method to improve the light extraction efficiency by embedding inverted MLA (IMLA) patterns with a feature size smaller than a few micrometers into an ultrathin substrate. It is not entirely clear how this small feature size, which is comparable to the wavelength of light, will affect EL emission. However, our results indicate that the proposed approach can lead to a significant enhancement in the external quantum efficiency (EQE) without the optical artifacts that can result from the diffraction effect or pixel blurring often found in devices with a thick MLA. Trans-scale optical simulation (BSDF) is then performed to reveal the mechanism suppressing the diffraction-induced artifacts.

Furthermore, we analyze the distribution of bending-induced stress and strain throughout the entire structure using the finite element method (FEM) to ensure the mechanical stability of the proposed system. By noting that the degree of the structural variation introduced by IMLA is comparable to the thickness of the substrate as well as the structural features specific to the given system, we establish an optimal mechanical design that minimizes the stress and strain applied to the transparent conductive oxide (TCO) electrode, which is the layer with the smallest COS within the device. With the proposed design, we demonstrate highly efficient foldable OLEDs exhibiting EQE as high as 58%. Furthermore, the devices are shown to retain ca. 90% of their initial brightness with virtually no drop in the EQE after 10,000 folding cycles at R_c as small as 50 μm.

RESULTS AND DISCUSSION

Inverted Microlens Arrays Embedded into Ultrathin Substrates. Figure 1a (left) presents a schematic illustration showing an ultrathin substrate based on UV-curable resin (NOA63, Norland Products, Inc.) containing IMLA patterns. The IMLA patterns are prepared from a mold, the surface of which has a normal convex MLA pattern based on an array of elliptical lenses with a height of 1.8 μm and a radius of 1.4 μm. Note that the height of the pattern is thinner than the thickness (10 μm) of the NOA63 substrate (hereafter “NOA substrate”) so that formation of the IMLA via engraving does not compromise the integrity of the substrate. The cross-sectional SEM image of the bottom NOA substrate shows that concave micrometer-sized patterns are well embedded into the

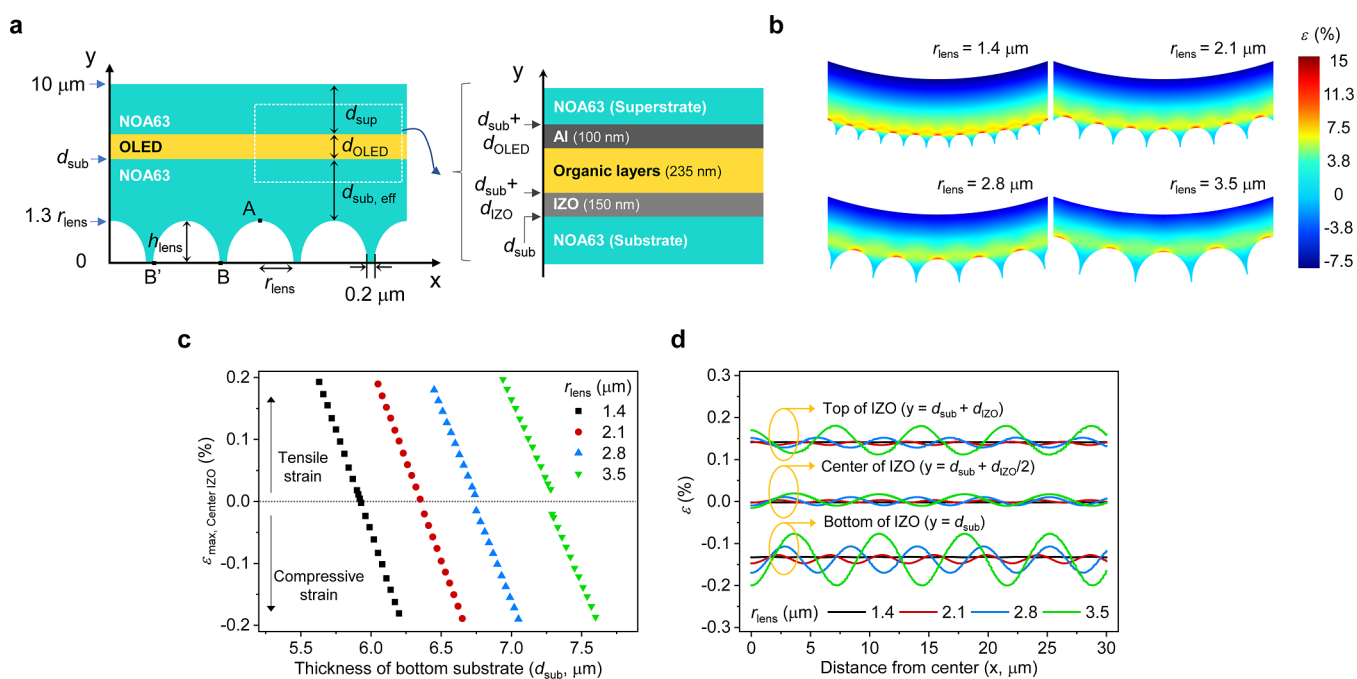


Figure 2. FEM-based 2D mechanical simulation of OLEDs fabricated on IMLA-embedded NOA substrates: (a) Overview of the device structure used in the FEM-based 2D mechanical simulation. The thickness of the entire device and the edge-to-edge distance between the lenses were fixed at 10 μm and 0.2 μm , respectively. In the elliptical lenses used in this work, the major axis (h_{lens}) was 1.3 times longer than the minor axis (r_{lens}). (b) Strain distribution of OLEDs with different lens sizes. (c) Maximum strain applied to the center of the IZO layer with different thicknesses of the bottom substrate (d_{sub}) and r_{lens} . Positive and negative strain values refer to tensile strain and compressive strain, respectively. (d) Strain distribution calculated for various r_{lens} values at the top, center, and bottom of the IZO layer vs position along the x direction.

substrate, as confirmed by the corresponding atomic force microscopic (AFM) image as well [Figure 1a (right, top)]. A detailed analysis of the height of the engraved MLA pattern from this AFM topology data indicates that an inverted MLA pattern feasibly replicates the MLA pattern of the mold with nearly identical dimensions (Figure S1).

As illustrated in Figure 1b, to form an ultrathin substrate with the embedded IMLA patterns, NOA63 optical resin was spin-coated onto the top of the mold with normal MLA patterns, with this step immediately followed by a UV curing process (Figure S2). The MLA mold was used as a carrier substrate during the manufacturing of the device, making it possible to minimize the complexity of the process and the handling steps that are associated with ultrathin substrates. Before the spin-coating of the NOA63 resin, a self-assembled monolayer (SAM) treatment was conducted with trichloro-(1H,1H,2H,2H-perfluorooctyl)silane (PFOCTS) on the MLA mold to prevent the fully cured NOA from sticking to the MLA mold and thus to realize easy mechanical exfoliation of the NOA substrate from the MLA mold when necessary (Figure S3). Once curing was done, an indium zinc oxide (IZO) layer was deposited as a TCO via radio-frequency (RF) sputtering on top of the proposed IMLA substrate, and organic/metal layers were subsequently deposited by thermal evaporation. The OLED stacks were based on Ir(dmpypy)₂tmd, which is a yellow phosphorescent emitter with high photoluminescence quantum yield and horizontal orientation of transition dipole moment (see Figure 1a for its molecular structure and the overall OLED stack used in this work).¹⁶ Once all the deposition steps were complete, the device was removed from the thermal evaporator and an additional NOA63 was spin-coated onto the top of the OLED stack in an N₂ environment as a top passivation layer. The OLED device

created in this way worked properly and exhibited excellent foldable characteristics, as shown in Figure 1a (right, bottom).

Mechanical Simulation of IMLA-Embedded Ultrathin OLEDs Using the Finite Element Method. Embedding a microscopic light extraction pattern into an ultrathin substrate can improve the optical efficiency, but its dimension being comparable to that of the substrate thickness can induce structural asymmetry between the upper and lower parts of the device. Thus, it is necessary to examine the mechanical characteristics of the entire device structure carefully to ensure good operation stability when the device is folded. In principle, it is critical to design an overall architecture that ensures the strain applied to layers vulnerable to folding-induced crack formation (e.g., TCO) to be sufficiently low so as not to cause a mechanical failure (see Figure S4 for an example). With this in mind, the strain distribution of each layer in the OLED under study was analyzed via a mechanical simulation based on the 2D FEM while the device was assumed to be bent at an R_c value of 50 μm . The device structure used in the simulation in each case consists of an IMLA-embedded NOA substrate/IZO (150 nm)/organic layer (235 nm)/Al (100 nm)/planar NOA superstrate with the thickness value of the thickest part being fixed at 10 μm , as depicted in Figure 2a.

Figure 2b presents the distribution of the strain for cases with different values of the short-axis length (r_{lens}) of an ellipsoidal concave lens in the IMLA. It should be noted initially that the strain is inhomogeneous along the horizontal axis, especially near the IMLA texture, and that it reaches its highest level near the point A, as shown in Figure 2a, that is, the vertex of each inverted microlens, while it is almost zero at its end tips (the points B and B'). The reason why the portions near the points B and B' exhibit almost zero strain and thus do not undergo any change in the local displacement is that they

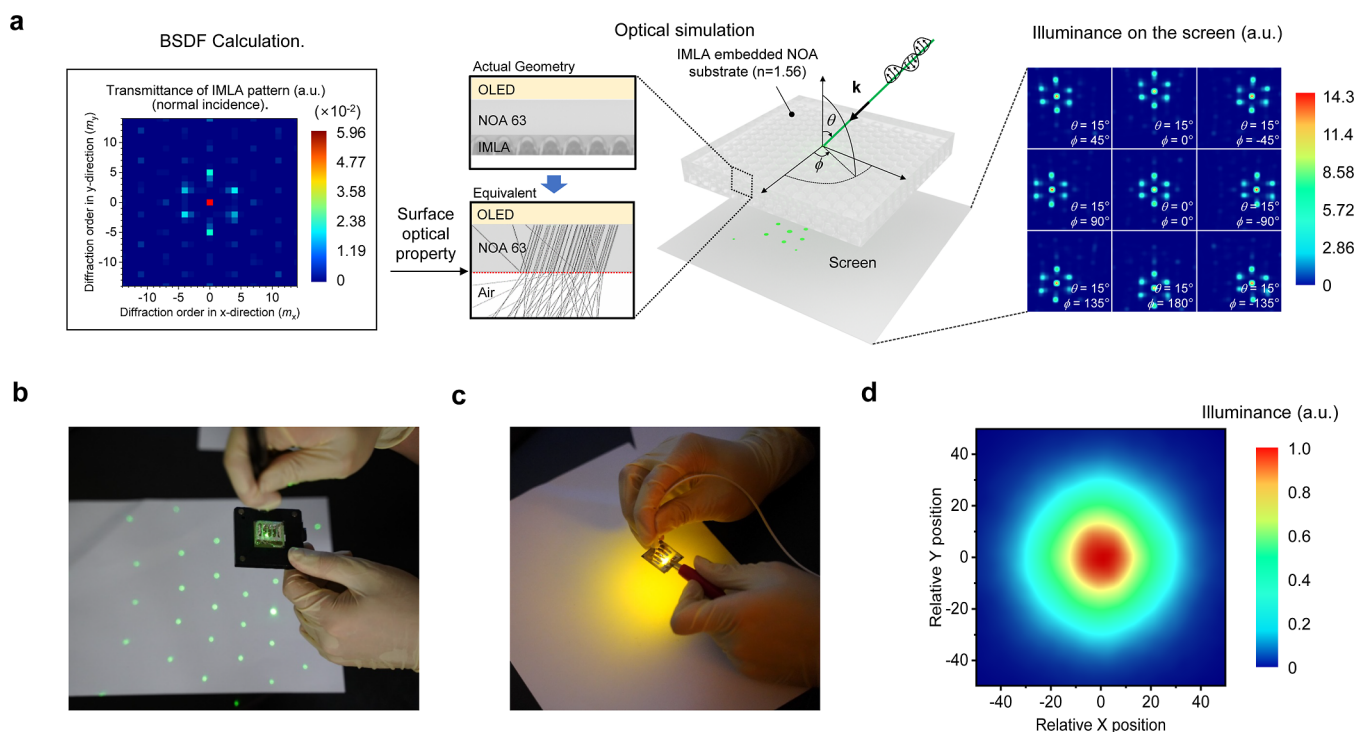


Figure 3. Optical effects of the IMLA pattern. (a) Overview of the optical simulation procedure. (Left) Example of the calculated transmittance distribution function for normal incidence obtained from BSGF calculation. (Middle) Inclusion of the BSGF result as a surface property of the NOA63/air interface in a ray-tracing software (LightTools) to set up an equivalent model of the proposed system. (Right) Output irradiance pattern on a screen for the light incident onto the IMLA surface from an OLED in a direction defined with polar and azimuthal angles θ and ϕ , respectively. (b) Hexagonally arranged diffraction pattern obtained when collimated light from a green-colored laser ($\lambda = 550$ nm) is passed through the IMLA pattern. (c) A photograph showing the light emission patterns of the fabricated OLEDs with the IMLA-embedded NOA. (d) Calculated irradiance patterns of the proposed OLED with the IMLA-embedded NOA.

are not connected to a continuous medium along the horizontal direction and can move essentially without restrictions in that direction. Instead, bending-induced stress is concentrated on the A points, yielding the largest strain therein and causing a spatial modulation that roughly follows the period of the IMLA, as shown in Figure 2b.

The IMLA-induced stress concentration effect and concomitant spatial variation of the strain make neutral plane engineering different from the conventional case. That is, for a typical case with both the substrate and superstrate showing planar similarity, the thicknesses of the substrate and the superstrate should be set nearly equal (i.e., $d_{\text{sub}} \approx d_{\text{sup}}$) to ensure that the thin layer sandwiched between them undergoes virtually zero stress upon bending. In the present case, however, $d_{\text{sub}} (\approx 10 \mu\text{m}) - d_{\text{sup}}$ leading to zero strain at the center of the IZO layer—the layer most vulnerable to bending-induced crack formation—is no longer equal to d_{sup} . For example, at $r_{\text{lens}} = 1.4 \mu\text{m}$, it is noted that setting d_{sub} to $5.92 \mu\text{m}$ and d_{sup} to approximately $4 \mu\text{m}$ is the proper choice for minimizing the strain or stress applied to the IZO layer. Likewise, when the r_{lens} values of an ellipsoidal lens are 2.1, 2.8, and $3.5 \mu\text{m}$, the optimal d_{sub} should be set to 6.35, 6.74, and $7.28 \mu\text{m}$, respectively, so that the IZO layer is positioned on the mechanical neutral plane (Figure 2c). Roughly speaking, it is found that engineering of the neutral plane in the present case should be done to let $d_{\text{sup}} \approx d_{\text{sub}}^{(\text{eff})}$, where $d_{\text{sub}}^{(\text{eff})} = d_{\text{sub}} - h_{\text{lens}} = d_{\text{sub}} - 1.3 r_{\text{lens}}$.

Furthermore, the simulation results reveal that designs based on IMLAs with small lenses are advantageous compared to those based on IMLAs with large lenses in terms of mechanical

flexural stability. For instance, Figure 2d indicates that when r_{lens} is relatively large, the strain at the top and bottom of the 150 nm thick IZO layer exhibits significant spatial variations even with a design that ensures zero strain at the center of the IZO layer. This result supports the validity of the present design that adopts an IMLA with a r_{lens} value as small as $1.4 \mu\text{m}$. It is also noteworthy that the present design with $r_{\text{lens}} = 1.4 \mu\text{m}$ allows a vertical range with sufficiently low strain (e.g., $\varepsilon < 0.2\%$) to be large enough not only to ensure low strain across the IZO layer and but also to allow for the future addition of an encapsulation structure consisting of oxide or nitride layers, which are also prone to bending-induced crack formation. It is noteworthy that the analysis shown here is for the cross section that passes through the centers (vertices) of the lenses in IMLA. When the simulation is done for other cross sections, the strain distribution could vary due to the different thickness distribution. Nevertheless, the variation in the strain across the IZO layer is small enough, with $r_{\text{lens}} = 1.4 \mu\text{m}$, to warrant that the strain applied to the IZO layer remains lower than its COS. (See Figure S5 in the Supporting Information for further details.) The variation in the thickness could also occur during the manufacturing. It is, therefore, important to check the design margin that can warrant that sufficiently low strain is applied across the IZO layer. The simulation result indicates that thickness variation by up to a few tens of percent is tolerable in the present case (Figure S6 in the Supporting Information).

Analysis of the Optical Effects of Embedded IMLA Patterns. Using a periodic structure with a spatial period of a few wavelengths is likely to induce a diffraction effect, which

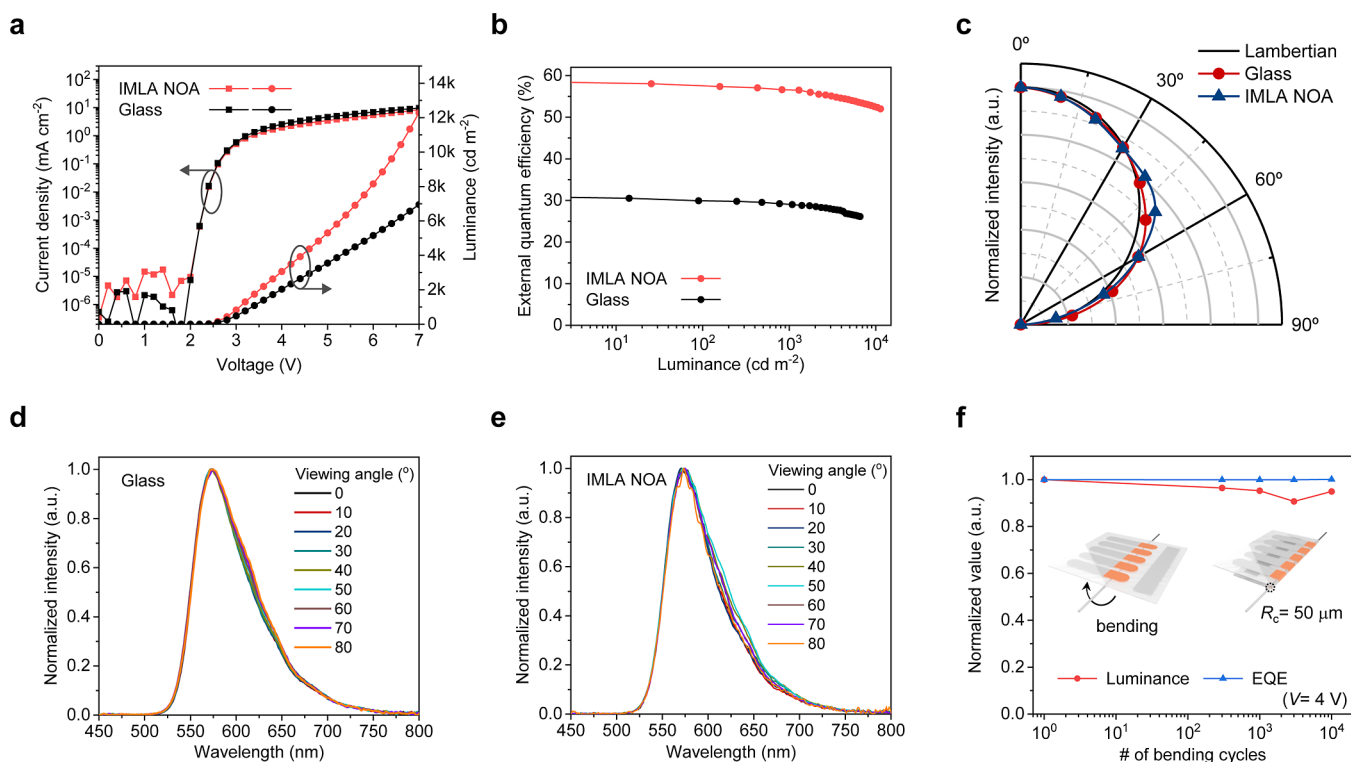


Figure 4. Device performance of OLEDs based on a glass substrate and the IMLA-embedded NOA substrate under study. (a) Current density and luminance versus the operation voltage. (b) External quantum efficiency versus luminance of the fabricated OLEDs. (c) Angular luminous intensity characteristics of the fabricated OLEDs. (d,e) Angular spectral intensity of the OLEDs fabricated onto (d) the glass substrate and (e) IMLA-embedded NOA substrate. (f) Normalized luminance and EQE versus number of bending cycles with a bending radius of 50 μm .

could cause optical artifacts in display applications. This is true for the present IMLA-based structure with $r_{\text{lens}} = 1.4 \mu\text{m}$, as indicated in Figure 3a,b, which illustrates that a hexagonal irradiance pattern can be produced when a collimated beam of light is irradiated in a direction perpendicular to the IMLA-embedded NOA substrate. Such a pattern is indeed the far-field (i.e., Fraunhofer) diffraction pattern expected from a 2D hexagonally arranged periodic structure. When working with an OLED in an integrated fashion, however, one should note that light from the OLED is incident to the IMLA not in a single, normal direction but in all different directions, which could be specified with polar and azimuthal angles θ and ϕ , respectively, as depicted in Figure 3a. To analyze its effect with regard to light coming internally from OLEDs in more detail, first we calculated the BSDF of the proposed IMLA structure using a diffraction analysis program called DiffractMOD (Synopsys, Inc.). The AFM data shown in Figure 1a were incorporated into the simulation to reflect the topographical data of the IMLA-embedded NOA substrate. Due to the hexagonal arrangement of the lenses in the IMLA, the hexagonal patterns were also observed in the transmission diagram (Figure 3a), which is natural considering that the diffraction pattern essentially follows the lattice in reciprocal space. This BSDF, that is, a collection of transmission/reflection data for incidence from all different directions, was incorporated into ray-tracing simulations by setting it as a surface property of the air/NOA interface. Subsequently, the angle-dependent intensity distribution of the OLED output to a substrate medium was obtained with the so-called “CPS” wave-optic model^{24,25} and assigned as that of the input source located right after the NOA/IZO interface. In this way, the net

output from an OLED passing through the IMLA-embedded NOA substrate was able to be simulated.

Even with the hexagonal pattern clearly visible for a plane wave incident in the normal direction, the output illuminance pattern from the OLED is found to exhibit cylindrical symmetry, which appears as if it were without any diffractive outcoupling structure (see Figure 3c,d for experimental and calculation results, respectively). Such beneficial characteristics stem from the fact that the light incident internally from the active layer of an OLED to its substrate has an angular distribution, which can be decomposed into a linear combination of plane waves with different wavevector directions. Considering that the far-field diffraction pattern follows the squared magnitude of Fourier transform of the incident field multiplied by the transmittance pattern, illuminance at the far field after passing through the IMLA for light with oblique incidence will have the identical hexagonal patterns but will be shifted from that of light with normal incidence, as illustrated for a few represented incident angles (the right side in Figure 3a). Because the light from the OLED exhibits cylindrical symmetry and is emitted in all directions, an incoherent addition of all hexagonally arranged illuminance patterns shifted from one another by varying displacements eventually leads to a cylindrically symmetric distribution without noticeable hexagonal characteristics, as can be confirmed in Figure 3c. This is the reason why the light output from an actual OLED fabricated on an IMLA-embedded substrate does show a smooth cylindrically symmetric pattern without visible artifacts (Figure 3d).

Enhanced Efficiency of OLEDs on the IMLA-Embedded Ultrathin Substrate. Based on the results discussed above, OLEDs were fabricated on the IMLA-embedded NOA

Table 1. Performance of the OLEDs with the IMLA Structures

device	external quantum efficiency (%)		current efficiency (cd A ⁻¹)		power efficiency (lm W ⁻¹)	
	max	@ 1000 cd m ⁻²	max	@ 1000 cd m ⁻²	max	@ 1000 cd m ⁻²
glass	30.5	29.0	85.4	81.1	117.2	78.6
IMLA NOA	58.0	56.5	165.3	160.9	222.7	162.6

substrate in the following configuration: IMLA-embedded NOA substrate (6 μm ; $r_{\text{lens}} = 1.4 \mu\text{m}$)/IZO (150 nm)/TAPC (130 nm)/TCTA (10 nm)/TCTA:B3PyMPM:Ir(dmppy-ph)₂tmd (30 nm, 4 wt %)/B3PyMPM (65 nm)/LiF (1 nm)/Al (100 nm)/Top NOA (3.5 μm), where TAPC, TCTA, Ir(dmppy-ph)₂tmd, and B3PyMPM refer to di-[4-(*N,N*-di-*p*-tolyl-amino)-phenyl]cyclohexane, 4,4',4''-tris(carbazol-9-yl)-triphenylamine, bis(2-(3,5-dimethylphenyl)-4-phenylpyridine) Ir(III) (2,2,6,6-tetramethylheptane-3,5-diketonate), and 4,6-bis(3,5-di(pyridin-3-yl)phenyl)-2-methylpyrimidine, respectively.¹⁶ It is noteworthy that the completed OLEDs show electrically stable operation even after detachment from the mold (Figure 4a). The maximum EQE outcomes of devices fabricated on glass and the IMLA-embedded substrate were measured and found to be 30.5 and 58.0%, respectively, demonstrating a 1.9 $\times$ enhancement from the already very high EQE of the control device (Figure 4b and Table 1). In fact, 58% is slightly higher than or comparable to the highest EQE value reported for OLEDs using the present emitter without a bulky half-ball lens or internal micro- or nanostructuring.¹⁶ While the previous high-efficiency result was realized with a thick volume scattering layer,¹⁶ the present results suitably illustrate the excellent potential of the proposed approach for efficiency enhancements compatible with ultrathin form factors. It should also be noted that there is little difference in the electroluminescence (EL) angular spectrum and that the angular distribution of the EL intensity is nearly identical to that of a Lambertian light source (Figure 4c–e). Furthermore, because the overall thickness of the proposed substrate with the outcoupling structure is sufficiently thin, the manufactured foldable OLEDs were found to be mostly free from pixel blurring, which is often regarded as the main obstacle preventing MLA and IMLA from being adopted in displays (Figures 1a and S7). The absence of pixel blurring means that the apparent emitting area is almost the same as the electrical active area; this is important in avoiding overestimation of luminance, which could occur when the outcoupling structure makes the effective emitting area much larger than that obtained without the outcoupling structure.

To assess the mechanical stability against repeated bending, a cyclic bending test was conducted using a string with a bending radius of 50 μm , as described by Lee et al.²⁶ As shown in Figure 4f, it was confirmed that the electrical characteristics and efficiency of the device did not deteriorate significantly even after 10,000 bending cycles in such a tight bending condition, demonstrating the validity of the mechanical design applied to the ultrathin OLEDs in this study.

CONCLUSIONS

In summary, we proposed a light outcoupling architecture for ultrathin OLEDs based on an IMLA embedded into an ultrathin substrate made of optical resin. With a resin coated over a mold having an array of convex ellipsoidal half microlenses with a radius (r_{lens}) of 1.4 μm and height ($h_{\text{lens}} = 1.3 r_{\text{lens}}$) of 1.8 μm , a highly flexible substrate with a thickness on the order of several micrometers and a refractive

index similar to that of glass was formed with the concave IMLA pattern embedded on one side. In addition, we carried out a FEM simulation to analyze the influence of the microstructure on the strain distribution throughout the device by noting that the dimension of the IMLA is comparable to the thickness of the substrate. With the mechanical impact of the IMLA taken into account, we identified, for various r_{lens} values, the thickness combinations of the IMLA-embedded substrate and a superstrate of the same optical resin that can lead to zero strain at the TCO layer (IZO in this work), which is most vulnerable to flexural crack formation. Also, embedding the IMLA pattern into the ultrathin substrate is more advantageous than forming MLA patterns outside of the substrate in terms of the mechanical stability. The modeling result revealed that embedding IMLA patterns with a small r_{lens} value is advantageous in that it ensures wide coverage of a low-strain zone. Micrometer-scale lenses arranged in a compact manner, however, could produce a diffraction pattern following the lattice geometry in the reciprocal space (hexagonal in the present case). To understand the effect of diffraction on the light output from an OLED, simulation was conducted by adopting a BSDF for the IMLA film. Simulation results illustrate that a plane-wave component incident on the IMLA does produce a hexagonal diffraction pattern; however, when one considers all the plane-wave components coming from the OLED in all different directions, a plurality of hexagonal diffraction patterns add up together with their centers shifted from one another by various degrees of displacement. This eventually smooths out the hexagonal characteristics and leads to an ideal, cylindrically symmetric light distribution with no visible artifacts.

The OLED device with the proposed IMLA structure showed nearly 1.9 times higher efficiency than the glass-based planar reference device, exhibiting a maximum EQE as high as 58%. With this IMLA structure, there was little optical blur owing to the ultrathin substrate. In addition, there was little difference in the EL angular spectrum, and the angular EL intensity was similar to that of a Lambertian light source. With the aforementioned neutral plane engineering tailored to the present system, the proposed OLED maintained its performance very well even under 10,000 repeated folding cycles at a bending radius as small as 50 μm . We believe the approach proposed in this study can shed light on a path forward in realizing highly efficient flexible light sources that are expected to play a major role in wearable or body-attachable optoelectronics for both display and biomedical applications.

METHODS

Preparation of the IMLA-Embedded NOA Substrates.

A sapphire substrate pre-patterned with the MLA structure (HNPS01, LUMTEC, Taiwan) was used for a mold. UV-curable polymer (NOA63 from Norland Optics) was used as an ultrathin substrate. For smooth mechanical exfoliation of the NOA substrate from the MLA mold, a self-assembled monolayer (trichloro(1H,1H,2H,2H-perfluorooctyl)silane, Sigma-Aldrich) was coated to form a hydrophobic surface on

the MLA mold. The MLA mold and a small vial containing SAM were placed in a desiccator, and the desiccator was pumped down with a rotary pump to evaporate the SAM. Thermal annealing was performed at 150° for 20 min to remove any residual SAM that did not adhere to the surface of the MLA mold. The thickness of the NOA substrate was controlled according to the rotational speed and time during the spin-coating process. The thickness of the NOA film was measured using a surface profiler (P-15, KLA-Tencor).

Optical Simulation. The AFM data of the MLA-embedded NOA were used as topological data for a diffraction analysis to predict the result through a BSDF calculation. In the BSDF calculation of the MLA mold, monochromatic case was assumed in order to shorten the simulation time. The simulation was performed by dividing the polar angle from 0 to 90° by 2.5° intervals and the azimuthal angle from 0 to 360° by 15° intervals. In a ray-tracing simulation (LightTools), the area of the OLED was set to 9 mm². A commercial software called DiffractMod was used to apply the diffraction effect of the IMLA pattern. Combining these BSDF results with the ray-tracing simulation, the final emission pattern was estimated with the angular distribution of light inside the substrate calculated with a full-wave optic model commonly referred to as the "CPS model."^{24,25}

FEM-Based Mechanical Simulation. The structural mechanics module of COMSOL Multiphysics was used for the numerical calculations of the spatial distribution of the stress and strain applied to the entire device. For the OLED devices, the Young's modulus of all organic layers in the device were assumed to be identical, and these layers were considered as one homogeneous organic layer. The Young's moduli of NOA63, IZO, organic layer, and Al were set to 1.655, 140, 2.75, and 70 GPa, respectively.¹² In the simulation, the bending situation was implemented by fixing a cylinder having a radius equal to the bending radius and then applying force to the edge of the device to bend it around the cylinder.

OLED Fabrication and Characterization. IZO layers were prepared on glass substrates and NOA substrates with a RF sputtering system (RF power, 120 W) in a vacuum (6×10^{-3} Torr). The temperature of the RF sputtering system was maintained at 50 °C during the sputtering process and during the 10 min post-annealing process. All substrates were treated by air plasma for 1 min in a plasma cleaner (CUTE-MP, Femto Science). Then, the substrates were transferred to a vacuum thermal evaporator to deposit an OLED stack at a pressure less than 5×10^{-7} Torr without breaking the vacuum. The OLED units under study had the following architecture: IZO/TAPC/TCTA/TCTA:B3PyMPM:Ir(dmppy-ph)2tmd (4 wt. %)/B3PyMPM/LiF/Al. TAPC, TCTA, and B3PyMPM were purchased from Nichem Fine Technology Co. Ltd. The phosphorescent emitter used here, Ir(dmppy-ph)2tmd, was purchased from Lumtec. The area of all devices was approximately 9 mm² and was estimated carefully with digitized Vernier calipers.

The current density, luminance, and electroluminescence (EL) spectra were measured using a source meter (Keithley 2400), a calibrated Si photodiode (FDS100-Cal, Thorlabs, Inc.), and a fiber optic spectrometer (EPP2000, StellarNet Inc.) in a nitrogen-filled glovebox, respectively. The photodiode and spectrometer were mounted on a custom-made goniometer to measure the angular EL spectra and intensity.²⁷ The EQE was calculated for the measured EL intensity and spectrum by fully considering the angular distribution.²⁷ The

bending test was carried out by repeating the operation of fixing the device around a rigid piano wire with a radius of 50 μm. Using a customized bending machine based on a motorized linear translator (T-LSR07SD, Zaber Technologies, Inc., Canada), the device went through a bending cycle in which the release state and the folded state progress alternately.²⁶

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsp Photonics.3c00017>.

AFM topology data of MLA patterns in MLA mold and IMLA patterns embedded in NOA substrates; surface morphology of the NOA 63 formed on the sapphire MLA mold; delamination process of the OLEDs from the carrier MLA mold; sheet resistance of 150 nm thick IZO film as a function of bending cycles with 1.4 and 1.8% of strain; strain distribution applied to the IZO layer according to the vertical positions at the IZO layer and the slicing paths; and maximum strain applied to the IZO layer with different thickness of the bottom substrate (PDF)

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Notes

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